

# Matched nulls and compact baselines qualify biochemical representations for metabolite reconstruction

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## Abstract

Biochemical groupings of metabolomics data are often justified by outperforming randomized controls, but that test does not establish practical predictive value. We evaluated representations by reconstructing hidden metabolites in held-out biological groups: 1,539 samples from 112 participants, 450 samples from 15 population categories and 876 cancer cell lines from 18 lineages. A qualification ladder compared each representation with matched nulls and with compact statistical predictors of equal size. Human nulls preserved descriptor sizes and lipid membership counts while rewiring which lipids co-occur; cell-line nulls matched neighbour counts, network degree and assay coverage. With median or fixed-neighbor aggregation, declared membership outperformed the expected matched null in every dataset (RMSE reductions of 0.100, 0.563 and 0.100 training standard deviations). Compact statistics nonetheless reduced error further: dimension-matched principal components by 38% and 32% in the human cohorts, and correlation-selected panels by 11% in cell lines; all five primary contrasts favored the statistical reference. In post-review analyses, replacing the median with an arithmetic mean closed 62–73% of the gap to principal components; in the larger cohort, mean aggregation showed no detectable advantage of declared over degree-matched rewired membership. Biochemical membership and aggregation should therefore be qualified separately, and against strong statistical baselines.

**Keywords:** benchmarking; lipidomics; metabolite imputation; pathway analysis; randomization

## Introduction

Metabolomics representations compress measured abundance profiles into quantities intended to be easier to compare and interpret. A useful evaluation must distinguish three questions: whether a representation predicts hidden measurements, whether its declared biochemical membership carries information beyond a matched random construction, and whether it competes with practical alternatives. These questions require different comparators, which can be arranged as a qualification ladder (Figure 1). A population mean tests basic predictive information. A matched null tests whether declared membership matters: it preserves each group's size and each metabolite's number of group memberships, but rewires which metabolites belong together. Pairwise overlaps between particular groups are not held fixed. An advantage over such a null therefore concerns which metabolites co-occur, not how many each group contains or how often each metabolite is used. A compact learned representation or

selected panel of the same size then tests whether the chosen construction is an effective way to use the available measurements. Passing the second rung does not imply passing the third.

The components of this approach have substantial prior art. Single-sample pathway scoring has been benchmarked in metabolomics [1], and pathway-local singular value decomposition was introduced through PLAGE [2]. LION/web organizes lipids by chemical and biological attributes [3]; subsequent LION-PCA work uses learned components to guide sample-level lipid summaries [4]. Thus, naming a lipid score by class, chain composition or ontology membership does not establish a new scoring method. It also does not establish that the named property is responsible for the score's predictive performance.

Metabolite reconstruction has similarly used network information and learned low-dimensional structure. MINMA combines metabolic-network and adduct relationships for missing-value prediction [5]. MIRTH reconstructs missing analytes using rank transformation and matrix factorization [6], while variational autoencoders have been evaluated for transferable metabolomics representations [7]. These methods address related but nonidentical endpoints: missing values, wholly unmeasured analytes, ranks or harmonized cohort representations. They establish relevant prior art without implying that a generic ridge or principal-component implementation constitutes a reproduction of their complete methods.

Null calibration is established in metabolic-network enrichment [8], and simulations have shown that pathway-analysis methods can recover nonspecific signals [9]. Our contribution is a reproducible empirical qualification of chemical and reaction-derived representations for hidden-metabolite reconstruction under biological-group exclusion, combining matched random constructions with strong compact predictors and separating the contribution of membership from that of aggregation. The analyses extend a previously released benchmark adaptively on cohorts whose earlier results were known; they qualify the tested implementations and do not rank biochemical against statistical methods in general.

## Results

### A qualification ladder separates membership information from competitive value

Each matched null preserves a stated structural budget of the biochemical representation and randomizes its biochemical content (Table 1). A null that is easier than this, for example random groups smaller than the declared ones, would be beaten for uninteresting reasons; one that also matched outcome correlation would be built from the answer. The primary human null is rebuilt inside every fit from the visible, training-eligible lipid-descriptor graph, so the preserved sizes and membership counts are those actually seen by each model. The CCLE null matches network degree and assay coverage but deliberately not correlation with the target. Throughout the ladder, matched nulls are summarized as expected values over 20 independent realizations, all under the current estimators and biological-group uncertainty.

| Null control                                 | Held fixed                                                                                       | Randomized                                                         | Used for                                                |
|----------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|
| Degree-preserving descriptor graph (primary) | Each descriptor's size and each lipid's number of descriptors, in the visible graph of every fit | Which lipids co-occur in a descriptor, and hence pairwise overlaps | Human ladder and aggregation analysis (20 realizations) |
| Property-matched random neighbours           | Number of neighbours per target; each neighbour's network degree and assay coverage              | Which metabolites serve as neighbours                              | CCLE ladder (20 realizations, current estimator)        |
| Size-matched random groups                   | Number and sizes of groups                                                                       | Which metabolites form each group                                  | Historical family scores (ST002081)                     |
| Dimension-matched random features            | Number of metabolite and expression features per target                                          | Which features are used                                            | Archived CCLE ensemble (released estimator; Supplement) |

Table 1. Matched null controls. Each preserves a stated structural budget and randomizes biochemical membership. Archived controls used the released estimator and are reported on that scale in the Supplement only.

With median or fixed-neighbor aggregation, both steps of the ladder held in every dataset (Figure 1E). Declared descriptor membership reduced error relative to the expected matched null by 0.100 training standard deviations in ST002081 (95% interval 0.095 to 0.105) and 0.563 in ST000818 (0.046 to 1.010); declared reaction neighbours reduced CCLE error by 0.100 (0.085 to 0.120). Every one of the 20 null realizations favoured declared membership in every dataset. Compact statistics then reduced error relative to the biochemical representation by 0.120 (0.115 to 0.126), 0.387 (0.075 to 0.687) and 0.092 (0.083 to 0.103), respectively.

A descriptive decomposition locates the ladder's gains. Between the training mean and the compact reference, the degree-matched rewired representation achieved 75% of the RMSE reduction in ST002081, declared membership a further 12% and the compact representation the remaining 14%; in ST000818 the shares were 19%, 48% and 33%, and in CCLE 25%, 39% and 36%. These shares depend on the loss scale: on the MSE scale the ST002081 shares become 88%, 7% and 6%. They are point-estimate partitions, not variance explained or a causal allocation of biological information. In particular, a rewired representation still averages biologically structured measurements and inherits each lipid's annotation-derived membership count, so it is not a no-information condition.

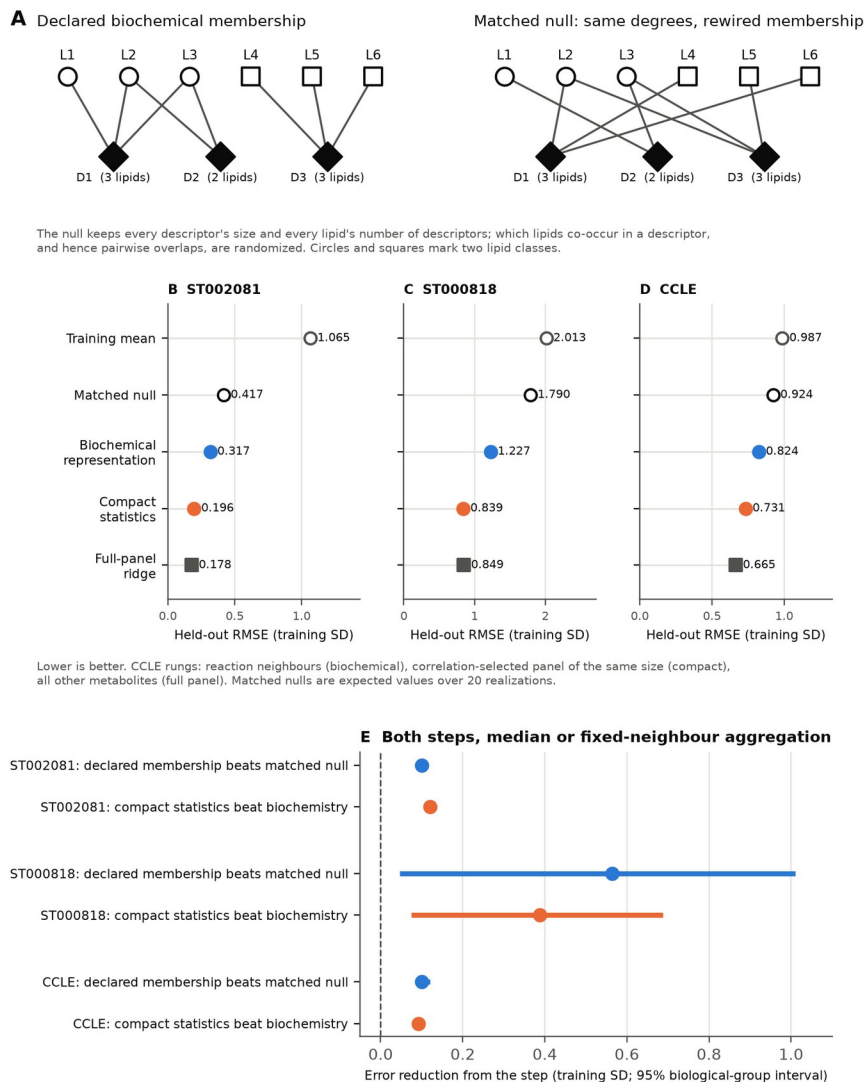


Figure 1. The qualification ladder. **A**, Construction of the primary matched null in a toy example: lipid-descriptor memberships are rewired so that each descriptor keeps its size and each lipid its number of descriptors, while which lipids co-occur, and therefore pairwise overlaps, change; biochemically coherent groups (circles or squares) become mixed. **B-D**, Held-out reconstruction error for each rung in ST002081 (112 participants), ST000818 (15 population categories) and CCLE (60 targets, 18 lineages), in training-standard-deviation units; lower is better. Matched nulls are expected values over 20 realizations. **E**, Error reduction contributed by the two steps with 95% biological-group bootstrap intervals (participants, population categories or lineages); intervals in ST002081 and for the CCLE null step are narrow. All estimates use the current estimators; the archived CCLE ensemble is reported separately in the Supplement.

## Compact statistics outperform the biochemical representations

At the same visible-input panel and representation dimension, global principal-component scores reduced error relative to descriptor medians from 0.3168 to 0.1964 in ST002081 and from 1.2267 to 0.8394 in ST000818, reductions of 38% and 32% (Table 2, Figure 2). In CCLE, selecting the same number of metabolites by absolute

correlation within the training data reduced error from 0.8239 for direct reaction neighbours to 0.7315, an 11% reduction. Both CCLE predictors used 1–19 metabolites per target, with median three. This comparison matches prediction-time marker count, not development information: correlation selection screened all 75 eligible non-target metabolites in the training set, whereas reaction neighbourhoods were fixed by prior annotation. No unselected held-out markers were used to form correlation-selected predictions.

The human panels supplied 390–397 visible lipid inputs per mask in ST002081 and 202–208 in ST000818. Structural median, local SVD and global PCA each produced 95 or 81 coordinates in the respective primary panels. All-visible ridge used the full visible dimension and achieved RMSE 0.178 and 0.849. Dimension matching therefore did not remove the reference methods’ advantages. It also did not equalize every development cost: PCA and local SVD learn loadings that the fixed median representation does not require.

| Dataset  | Descriptor median | Descriptor mean* | Descriptor-local SVD | Global PCA | All-visible ridge | Matched null (expected) |
|----------|-------------------|------------------|----------------------|------------|-------------------|-------------------------|
| ST002081 | 0.3168            | 0.2425           | 0.2434               | 0.1964     | 0.1780            | 0.4168                  |
| ST000818 | 1.2267            | 0.9443           | 0.9574               | 0.8394     | 0.8491            | 1.7901                  |

Table 2. Human primary results: square root of equal biological-group MSE in training-target standard-deviation units. Groups are 112 participants or 15 population categories. Models share their cohort’s outer splits and within-fit eligibility. Matched nulls throughout Tables 2 and 4 and Figure 1 are expected values over 20 realizations of the degree-preserving null; the single primary realization gave 0.4192 and 1.8030, and is the null shown in Figure 2. \*Descriptor mean is a secondary, outcome-aware aggregator on identical memberships and is not part of the five primary contrasts.

An all-other-metabolite model reached 0.6648. Global PCA with the same number of coordinates as the direct panel reached 0.8090, while reading all 75 non-target assays at prediction time. Its coordinate count therefore must not be described as a matched assay budget. Adding reaction-linked expression signatures yielded 0.8117; adding metabolite-expression products yielded 0.8170. The interaction construction did not improve the aggregate point estimate over the additive model, preserving the historical absence of a general interaction gain. These ablations were descriptive rather than extra primary significance tests.

The additive and interaction models consumed additional expression information: median five unique genes and five expression signatures, with median representation dimensions eight and thirteen. A low-dimensional output is therefore an incomplete description of information cost. The full budget records expose metabolite inputs, raw transcript genes, signature counts, interaction terms, loadings and regression coefficients for each outer fit.

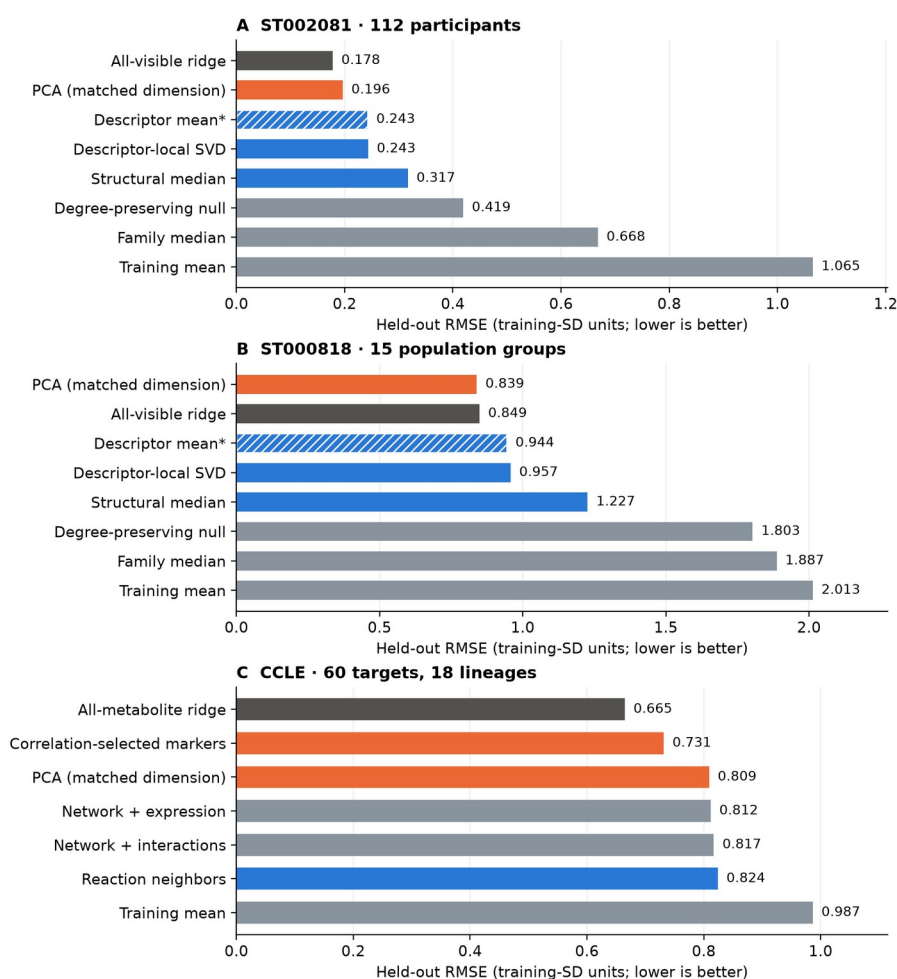


Figure 2. Reconstruction performance in the accepted adaptive comparisons. Lower RMSE indicates better prediction. Human scores weight participants or population categories equally; CCLE scores average available lineages within each fixed target, then targets. Dataset-specific training-standardized units and different target panels preclude interpreting differences between datasets as a common biological effect size. The degree-preserving null bar is the single primary realization (Table 2 gives the 20-realization expected values). Descriptor mean (hatched) is a secondary, outcome-aware aggregator, not part of the five primary contrasts; descriptor-local SVD is shown with the biochemically structured models because it learns weights within declared descriptors. All-model and sensitivity tables retain the training-mean, family and null references.

## The five declared contrasts favour their statistical references

All five prespecified contrasts favoured the statistical reference (Table 3, Figure 3). We defined improvement as reference RMSE minus the structural model's RMSE, so negative values favour the reference. The 95% descriptive intervals and the 99% per-contrast intervals, which target nominal 95% familywise coverage across the five contrasts, excluded zero. The contrasts reuse cohorts, fitted-analysis structures and, in the human comparisons, the same median reference; they are five declared comparisons, not five independent replications. Their intervals are conditional on the fitted predictions and fixed target panels, and overlapping cross-validation training sets

induce dependence that the resampling does not fully represent (Methods). The direction and size of the gaps, rather than the resolution-limited sign-flip probabilities, are therefore the primary evidence.

| Dataset and reference            | Difference | 95% interval       | 99% per-contrast interval |
|----------------------------------|------------|--------------------|---------------------------|
| ST002081, global PCA             | -0.1204    | -0.1258 to -0.1153 | -0.1278 to -0.1140        |
| ST002081, local SVD              | -0.0733    | -0.0776 to -0.0691 | -0.0789 to -0.0676        |
| ST000818, global PCA             | -0.3873    | -0.6872 to -0.0748 | -0.7902 to -0.0712        |
| ST000818, local SVD              | -0.2693    | -0.4725 to -0.0518 | -0.5388 to -0.0494        |
| CCLE, correlation-selected ridge | -0.0924    | -0.1034 to -0.0830 | -0.1073 to -0.0807        |

Table 3. Paired contrasts from 10,000 biological-group bootstrap draws. Difference is reference RMSE minus structural-model RMSE; negative values favour the reference. The 99% per-contrast intervals target nominal 95% Bonferroni familywise coverage across five contrasts, subject to the conditional bootstrap assumptions. The structural model is descriptor median for human datasets and direct-neighbor ridge for CCLE. Intervals use participants, population categories or lineages; metabolites and validation folds are not counted as independent people.

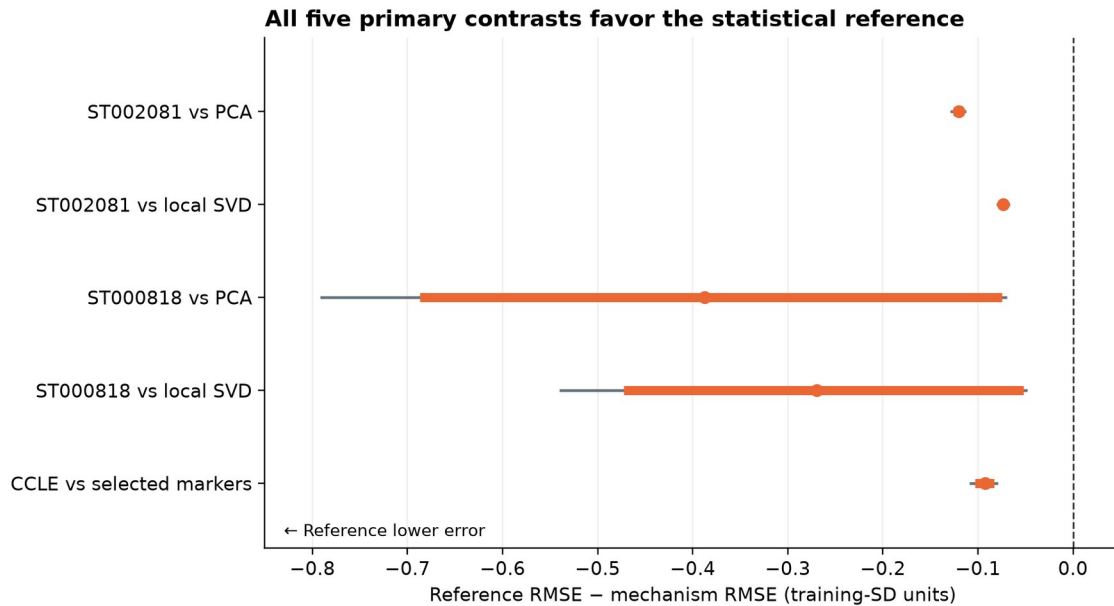


Figure 3. Primary reference-minus-structural-model RMSE contrasts. Zero denotes equal error; negative estimates favor the reference. Thick intervals are descriptive 95% intervals; thin intervals are 99% per-contrast intervals targeting nominal 95% Bonferroni familywise coverage across five contrasts. Resampling preserves paired outcomes and, for CCLE, the fixed target panel. It does not refit models or characterize uncertainty from choosing a different cohort, assay panel or annotation database.

### The membership advantage depends on aggregation

Aggregation strongly affected accuracy. Replacing the median with an unweighted arithmetic mean on identical memberships reduced error to 0.2425 in ST002081 and 0.9443 in ST000818, comparable to descriptor-local SVD (0.2434 and 0.9574) and

closing 62% and 73% of the median-to-PCA gap without learning within-descriptor weights (Table 2). The mean was requested after the primary outcomes were known and is reported as secondary.

We therefore crossed declared versus rewired membership with median, mean and local-SVD aggregation in a post-review analysis that reused the primary splits and tuning and averaged over 20 null realizations (Table 4, Figure 4). In ST000818, declared membership outperformed its matched null under all three aggregations. In ST002081 it did so for the median (0.100) and local SVD (0.035), but not for the mean: under mean aggregation, the sampled degree-matched rewired representations achieved nearly identical aggregate error to the declared descriptors (difference  $-0.001$ ; 95% interval  $-0.003$  to  $0.001$ ; positive in 5 of 20 realizations). This is an absence of detected advantage across 20 sampled rewirings, not a demonstration of equivalence for every permissible graph. In the larger cohort, the apparent membership advantage of the median therefore depended on the aggregation rule. Biochemical membership added information conditionally on how members were combined, and global covariance captured by PCA remained more informative than any descriptor-based aggregation tested.

| Dataset  | Rule         | Declared | Null<br>(expected) | Effect (95%<br>interval)            | Realizations<br>favouring<br>declared |
|----------|--------------|----------|--------------------|-------------------------------------|---------------------------------------|
| ST002081 | Median       | 0.3168   | 0.4168             | 0.100 (0.095 to 0.105)              | 20/20                                 |
| ST002081 | Mean*        | 0.2425   | 0.2417             | $-0.001$<br>( $-0.003$ to $0.001$ ) | 5/20                                  |
| ST002081 | Local<br>SVD | 0.2434   | 0.2789             | 0.035 (0.033 to 0.038)              | 20/20                                 |
| ST000818 | Median       | 1.2267   | 1.7901             | 0.563 (0.046 to 1.010)              | 20/20                                 |
| ST000818 | Mean*        | 0.9443   | 1.0592             | 0.115 (0.012 to 0.216)              | 20/20                                 |
| ST000818 | Local<br>SVD | 0.9574   | 1.3620             | 0.405 (0.019 to 0.764)              | 20/20                                 |

Table 4. Declared versus rewired membership by aggregation (post-review, outcome-aware). Rule is the aggregation rule. Effect is expected-null RMSE minus declared RMSE; positive values favour declared membership. Intervals are paired biological-group bootstrap intervals (10,000 draws). \*Secondary aggregator.

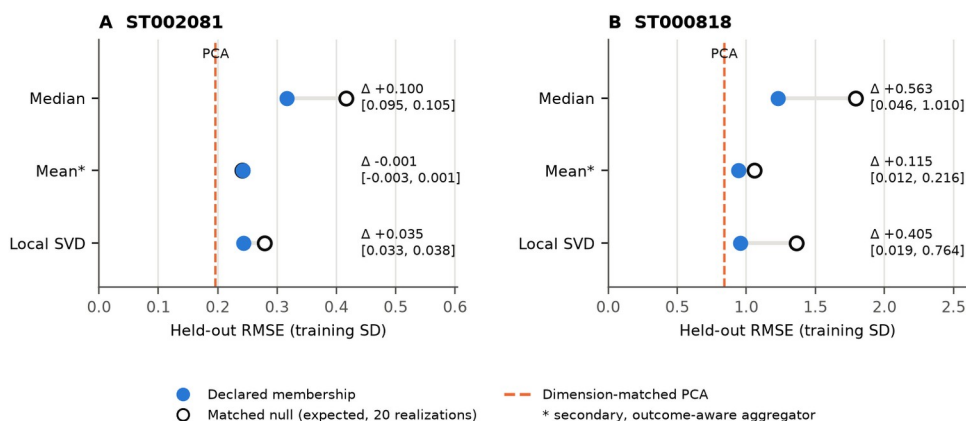


Figure 4. Membership effect by aggregation rule. Held-out RMSE for declared descriptor membership (filled) and the expected degree-matched null over 20 realizations (open) under median, mean and descriptor-local SVD aggregation in **A**, ST002081 and **B**, ST000818.  $\Delta$  is the null-minus-declared RMSE with its 95% biological-group bootstrap interval; the dashed line marks dimension-matched PCA. The mean is a secondary, outcome-aware aggregator; the whole crossing is a post-review analysis.

## Sensitivities preserve the principal accuracy ordering

Training-defined missingness eligibility expanded the ST002081 visible panels to 571–596 inputs. Structural median remained less accurate than PCA and local SVD: RMSE 0.310, 0.214 and 0.241 respectively. ST000818's missingness run gave the same aggregate results as its primary panel. This sensitivity reduces dependence on whole-cohort complete-case selection; it does not turn the reused cohort into unseen confirmation.

A separate annotation-resolution sensitivity removed unsupported individual-chain descriptors from multi-chain sum-composition labels while retaining declared total composition. ST002081 results changed little. In ST000818 the representation shrank from 81 to 67 coordinates; median, PCA and local-SVD errors were 1.235, 0.863 and 0.962. Thus neither sensitivity reversed the principal ordering. These checks qualify the implemented annotations rather than validating every molecular identity.

The earlier released benchmark was also reproduced. Broad-family scores in ST002081 did not beat size-matched random groups under the archived estimator (row-weighted RMSE 0.650 versus 0.627), whereas the archived descriptor and CCLE structural analyses favoured their declared constructions over the corresponding null ensembles; the archived CCLE property-matched advantage was 0.137 (target-bootstrap 95% interval 0.097 to 0.181). These archived values use the released estimator and are reported in the Supplement rather than combined with the current ladder.

## Discussion

Beating a matched randomized construction is not enough to establish practical predictive value. In every dataset, declared biochemical membership with median or fixed-neighbour aggregation outperformed matched nulls, yet compact statistical predictors of the same size reduced error by a further 11–38%. The project's original

predictive-superiority ambition was not met, and a general superiority claim for the tested biochemical representations is unsupported.

The aggregation results sharpen this conclusion. Much of the median's disadvantage disappeared with a simple arithmetic mean, and in the larger cohort mean aggregation showed no detectable advantage of declared over rewired membership. Membership choice, aggregation rule and access to global covariance are therefore separate contributions. Descriptor-local SVD, which learns weights within declared descriptors, is better regarded as a biochemically structured statistical representation than as a pure test of biochemistry. We suggest that studies proposing biochemical representations report three things: the gain over a null that matches the representation's structural budget, the sensitivity of that gain to the aggregation rule, and the gap to a compact statistical alternative of the same size. Any one alone invites a misleading conclusion.

The evidence has important boundaries. Both human cohorts and their archived outcomes were known before the extension. Historical complete-case panels, CCLE mappings and lineage-size eligibility were retained to make comparisons auditable. The missingness sensitivity addresses one selection mechanism, while assay coverage, lipid identification and network accessibility still limit scope. Training weights each sample equally, so frequently sampled people contribute more to model fitting, even though the primary evaluation weights people equally. Fifteen population categories and 18 cancer lineages provide limited support for inference to substantially different populations or tissues. The task hides selected analytes within existing assay panels; it does not establish cross-platform transfer, reconstruction of entirely absent biochemical classes or value for other downstream endpoints. Uncertainty is conditional on fixed fits; performance across alternative group partitions with full refitting, and an untouched cohort, would provide stronger tests.

Related methods were treated as prior art, not as unperformed benchmark wins. The local SVD is an implementation of a standard principle, not a full ssPA package evaluation. Neither MINMA, MIRTH, LION-PCA nor a deep representation was executed as a native comparator. MIRTH code was excluded because applicable commercial-research and publication permissions were unresolved. The present result concerns the tested methods and endpoint; it establishes no ranking against those systems.

The broader factorized evidence programme from which this benchmark derives (genetic, constraint-based and perturbational components) was not newly validated here. Concentration reconstruction does not establish flux, temporal causality, treatment response or clinical utility.

## Methods

### Data, source boundaries and eligibility

The human discovery data were ST002081/AN003790, associated with Hornburg et al. [10]; the evaluated matrix contained 1,539 repeated samples and 112 participant codes, matching the final publication (the Workbench narrative's 1,546 samples/109 subjects is a documented metadata discrepancy). The second dataset, ST000818/AN001299, contained 450 samples across 15 population categories [11]. Cohorts were analysed separately, with models refitted within each cohort. CCLE metabolomics and expression provided 913 aligned cell lines [12]; the frozen mapping

retained 76 of 225 assayed metabolites, and eligibility yielded 60 targets evaluated on 876 lines from 18 lineages. Direct-reaction features used Human-GEM v2.0.0 [13,14].

The earlier public software [15] was re-executed in a fresh environment: 51 of 62 declared outputs of 14 real-data stages were byte-identical, the recorded analysis decisions agreed, and remaining numerical differences are retained in the Supplement. An initial human extension implementation was superseded after defects in inner-fold weighting, rank capping, target eligibility and loss aggregation were found; its outputs are archived and excluded. The comparator set, split rules, tuning grid, estimands and five primary contrasts were retained through these repairs.

Inputs were recovered into an isolated workspace with source checksums. ST002081 used the pinned Workbench AN003790 table: 1,643 rows, of which 104 QC rows had RandomID = NA; 1,539 rows and 112 participant codes remained. The archived primary schema contained 493 lipids in nine families. ST000818 used AN001299 and its Categorization factor, yielding 450 sample rows, 15 groups and 255 primary lipids in six families. Unique sample labels and the deposit's 450-person statement were retained separately from the unavailable donor-level crosswalk. Abundance values were used in the provided scale before training-fold standardization; no additional human log transformation was introduced.

CCLE input files were CCLE\_metabolomics\_20190502.csv, CCLE\_RNAseq\_genes\_rpkms\_20180929.gct.gz and Cell\_lines\_annotations\_20181226.txt. The loader retained source-provided cleaned log10 metabolite abundance and transformed expression to  $\log_2(\text{RPKM} + 1)$ , averaging duplicate gene rows before transformation. The 913 aligned lines supplied 76 mapped metabolites. Frozen candidate definitions required direct non-transport reactions containing at most eight metabolites and complete reaction-linked gene signatures. Sixty target definitions were reconstructed from 317 candidate rows. For each target, finite target measurements and lineages with at least 15 eligible lines were required; target eligibility also required at least 300 lines and five lineages. All 60 accepted targets retained 876 lines and 18 lineages. These restrictions were applied retrospectively to the known assay and are disclosed as such.

Human-GEM v2.0.0 supplied reaction and gene-protein-reaction mappings. Alternative GPR disjuncts were represented separately; multi-gene complex signatures used the minimum constituent log-expression value, with single-gene signatures unchanged. This is an expression proxy, not an enzyme-activity measurement or flux constraint calibration. Mapping/candidate reconstruction was compared with the frozen files. The input ledger, rejected mappings and per-target eligibility accompany the reproducibility dossier.

## Human representations and hidden panels

Five disjoint masks distributed eligible features within families, so each primary lipid was hidden once. Hidden analytes were removed before fitting every representation. Label-derived descriptors covered family, total carbon/unsaturation, family-by-unsaturation and parsed chain attributes. Descriptors had at least five panel members and at most 90% panel coverage; after masking and training eligibility, at least two visible members were required. Median scores and local first-component scores used identical memberships. Global PCA used the same visible panel and a component count equal to retained descriptor count, capped by numerical rank of the centered training matrix. Rank used the largest singular value times the largest matrix dimension times floating-point epsilon as tolerance.

Family medians and all-visible ridge were retained. The primary randomized descriptor graph was rebuilt inside every fit from the visible, training-eligible lipid-descriptor incidence after masking and the 80% availability rule. Bipartite edge switches, ten attempts per edge, were accepted only when they kept a simple graph; the code asserts exactly preserved descriptor sizes and lipid membership counts in each fit. Pairwise descriptor overlaps, and therefore which lipids co-occur, are not preserved. Edge switching with these mixing settings does not guarantee uniform sampling over all degree-matched graphs.

The primary schemas inherit whole-cohort complete-case selection. In the missingness sensitivity, all assay labels belonging to families of at least five labels were candidates. Every fit retained visible inputs observed in at least 80% of its training rows. Hidden targets had to be complete and nonconstant within that fitting partition; validation losses used only observed targets. Inner eligibility was recalculated from inner-training rows, independently of outer and inner validation missingness. The resolution sensitivity removed acyl, chain-carbon and chain-unsaturation assignments from multi-chain labels containing only a total-composition pair; single-chain and explicitly reported chain annotations were retained. Both sensitivities used all original models and unchanged tuning rules.

## CCLE predictors and information budgets

For each target, direct-neighbor ridge used its frozen set of  $k$  measured neighbors. Correlation-selected ridge ranked all 75 non-target training metabolites by absolute Pearson correlation with the training response after mean imputation, selecting  $k$  with stable lexical tie handling. Selection was repeated within every inner and outer training partition. Prediction used only those selected markers. Global PCA read all 75 non-target metabolites, with at most  $k$  coordinates and a numerical training-rank cap. All-other-metabolite ridge used the full panel. Additive models combined neighbors and GPR signatures; interaction models additionally included mapped products of standardized metabolite and expression terms, standardized again from training moments. The target was excluded explicitly from predictor matrices and interaction definitions.

Prediction-time metabolite count, candidate-screening count, transcript genes, signature/interaction dimensions, loadings and regression coefficients were recorded separately. Human PCA and descriptor scores match visible inputs and output coordinates, not learned-transform parameters. CCLE selected panels match prediction-time metabolite count while requiring a broader training assay. Global CCLE PCA matches coordinates only.

## Training, validation and estimands

Each input column was imputed with its arithmetic training mean, then divided by the population standard deviation of the imputed training column after centering. Empty-column means defaulted to zero and scales at or below  $10^{-12}$  to one. Representations, including PCA component scores, were standardized from their own training moments before ridge. Targets were standardized using training means and population standard deviations; held-out target values never supplied a transform or predictor. The training-mean reference predicted zero in these units. Regressions included an intercept and minimized squared error plus  $\alpha \parallel \beta \parallel_2^2$ .

Human outer validation used one balanced five-fold group partition shared across all masks/models. Groups were participants for ST002081 and population categories for ST000818. CCLE used two repeats of five lineage-isolated folds per target. Three grouped inner folds selected  $\alpha \in \{0.1, 1, 10, 100\}$  by pooling individual validation-group MSEs with equal group weight; exact ties favored the smaller value. Imputation, scaling, loadings and supervised marker selection were refitted within each inner training set, then on outer training using the selected alpha. Human regression used the Cholesky solver; CCLE used LSQR. Training rows were equally weighted. Split seeds and complete tuning grids are recorded with each run.

For human data, let  $e_{im}$  be the mean squared standardized error over observed hidden targets for sample  $i$ , mask  $m$ . Average nonempty masks within a sample, then samples within group  $g$ , to obtain  $L_g$ . The primary error is

$$RMSE_{human} = \sqrt{\frac{1}{G} \sum_{g=1}^G L_g}.$$

Repeated samples stay within their participant. For CCLE, let  $L_{t\ell}$  average squared errors over cells and repeated predictions for target  $t$ , lineage  $\ell$ . With fixed  $T=60$ , available lineages  $A_t$  and their number  $n_t$ ,

$$RMSE_{CCLE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \frac{1}{n_t} \sum_{\ell \in A_t} L_{t\ell}}.$$

CCLE repeats are averaged as losses, not counted as new cells. Supplementary row-weighted RMSE pools squared errors over observed prediction pairs. The supplementary skill statistic  $1 - \sum_j (z_j - \hat{z}_j)^2 / \sum_j z_j^2$  uses standardized targets  $z_j$  and predictions  $\hat{z}_j$ , with targets centered by training-fold means; it is improvement over that reference, not conventional test-set-centered  $R^2$ .

## Qualification ladder and post-review analyses

Ladder rungs are the primary equal-group or equal-lineage RMSEs of the training mean, the expected matched null, the declared representation (descriptor median or direct reaction neighbours), the compact reference (dimension-matched PCA or correlation-selected panel) and full-panel ridge. Compact steps are the locked primary contrasts. Shares divide the training-mean-to-compact error reduction into successive differences on the RMSE scale, with MSE-scale shares reported for comparison; they are point-estimate partitions without intervals.

Two post-review, outcome-aware analyses were frozen in a written amendment before their outcomes were computed and do not alter the five primary contrasts. First, for the human cohorts, declared and rewired memberships were each combined by median, arithmetic mean and descriptor-local SVD, reusing the primary code, splits and tuning; null realization  $k$  used seed (primary seed + 1000k), with  $k = 0$  reproducing the primary null exactly, and the same null graph served all three aggregations within a fit. Second, for CCLE, property-matched random neighbour sets were generated with the archived assignment algorithm (network degree and assay coverage; seeds 20262100–20262119) and fitted with the accepted direct-neighbour ridge on the accepted outer splits. Expected nulls average group- or target-lineage MSE across the 20 realizations

before comparison. Membership effects are expected-null RMSE minus declared RMSE, with paired biological-group bootstrap intervals (10,000 draws, seed 20260921). The archived CCLE ensemble, which used the released estimator and target-level resampling, is retained only in the Supplement. The toy example in Figure 1A preserves both degree sequences, which the figure script asserts.

## AI-assisted research and writing

OpenAI Codex and Anthropic Claude were used under the author's direction for code drafting, analysis orchestration, literature organization, figure preparation and manuscript-language development. The author defined the scientific questions and claim boundaries, reviewed the analysis code and outputs, verified citations and numerical claims, revised the manuscript and accepts full responsibility for its content. AI systems were not treated as authors or as independent sources of evidence.

## Uncertainty, adaptation and reproducibility

The five primary contrasts were descriptor median versus global PCA and local SVD in each human cohort, plus CCLE direct neighbors versus correlation selection. Improvement equals reference RMSE minus structural-model RMSE. Paired bootstrap draws resampled 112 participants, 15 population categories or shared CCLE lineage labels, preserving all fixed targets and repeated losses. Ten thousand draws used seed 20260921. Percentile 95% intervals were descriptive; 99% per-contrast intervals allocated 0.05 across five contrasts by Bonferroni, targeting nominal 95% familywise coverage. Two-sided paired sign flips used seed 20260922, 10,000 draws and a plus-one Monte Carlo correction; their five probabilities received Holm adjustment. All five raw probabilities reached the resolution floor of 1/10,001. For CCLE, each lineage's paired MSE contribution was flipped jointly across targets. Sign flips assume exchangeability of paired group-effect signs under the null. Overlapping cross-validation training sets can induce dependence across held-out groups that neither procedure fully represents. Uncertainty is conditional on fixed fits, targets and these resampling assumptions, with no full-pipeline refitting or population-sampling guarantee.

The specification and resolution amendment were frozen before extension outcomes, after earlier cohort results had been inspected. The work is therefore post-release/adaptive, not prospective or blind. An outcome-aware descriptor-mean follow-up requested after the primary results, and the post-review analyses above, are reported as secondary and descriptive; they are not part of the five-test family. Superseded human implementation outputs are explicitly excluded. Accepted human revision 2 and CCLE implementations have distinct code hashes, input checks, split/eligibility records and meaningful numerical tests. The environment used Python 3.12.3, NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.1 and scikit-learn 1.9.0, with single-threaded BLAS and bounded local concurrency. Historical implementation-fingerprint discrepancies remain disclosed; no tolerance was chosen to relabel differing files as byte-identical.

## Data availability

ST002081 and ST000818 are available from the Metabolomics Workbench (<https://doi.org/10.21228/M8ZM5P> and <https://doi.org/10.21228/M89M31>) under CC BY 4.0. CCLE metabolomics, expression and annotation files are available from the original release [12] and are not redistributed. Human-GEM v2.0.0 is available under CC BY 4.0

[13,14]. Aggregate results, figure source data, run manifests and checksums are included in the code archive. Participant-linked losses and split maps are not redistributed.

## Code availability

The qualification-ladder workflow, extension analyses, tests and figure code are openly available under the Apache-2.0 licence at <https://github.com/otcan/metabolic-representation-qualification> (release v1.1.0). The exact release is archived at Zenodo (<https://doi.org/10.5281/zenodo.23100294>). The earlier matched-null software is available as release v1.0.1 [15]. Both run on standard Python and are free of charge for academic and non-academic use.

## Ethical statement

**Acknowledgments:** Computational resources were provided by Metastate Bio Inc.

**Research ethics:** Not applicable. This study reanalysed publicly available, de-identified data and involved no new recruitment, intervention or data collection; ethical approval for the source studies is described in the original publications and deposits [10,11].

**Informed consent:** Not applicable.

**Author contributions:** The author has accepted responsibility for the entire content of this manuscript and approved its submission. O.Ü. conceived the study, defined the research questions and claim boundaries, directed the computational programme, provided resources, interpreted the results and revised the manuscript.

**Use of Large Language Models, AI and Machine Learning Tools:** OpenAI Codex and Anthropic Claude were used under the author's direction for code drafting, analysis orchestration, literature organization, figure preparation and manuscript-language development, as described in the Methods. The author verified all code outputs, citations and numerical claims.

**Conflict of interest:** O.Ü. is the founder of Metastate and is affiliated with Metastate Bio Inc, which develops commercial computational biology, biomarker and modelling products and services that could benefit from the publication of this work. The author declares no other conflict of interest.

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**Data availability:** See the Data availability and Code availability sections.

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